Rounak Ray

Curriculum Vitáe

Office 310, Division of Applied Mathematics, Brown University
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Research Interests

Random graphs, Statistical physics, Interacting particle systems, Random walks, Random interface models, Large deviation

Work Experience

2025 - 2027 Division of Applied Mathematics, Brown University Post-doctoral research fellow

Jan - May 2025 SLMath (formerly **MSRI**) Post-doctoral research fellow

Education

2020–2024 **PhD**, *Eindhoven University of Technology*.
Department: Mathematics & Computer Science
Supervisors: Remco van der Hofstad, Rajat Subhra Hazra
Dissertation: Stochastic Processes on Preferential Attachment Models

2018–2020 **Master of Statistics (Hons.)**, *Indian Statistical Institute, Kolkata*.
Specialisation: Probability Theory

2015–2018 **Bachelor of Statistics (Hons.)**, *Indian Statistical Institute, Kolkata*.

Professional Experience

November 2023 Organizer of "Lecture series on interacting particle systems on random graphs"

2021–2023 Teaching Assistant, Eindhoven University of Technology: Longitudinal Data Analysis

2022–2024 Teaching Assistant, Eindhoven University of Technology: Random Graphs

Publications

2026 S. Bhamidi, F. den Hollander, R. van der Hofstad, R. Ray **Network evolution with self-reiforcement**. ArXiv Preprint

2026 R. van der Hofstad, R. Ray **Ising model on preferential attachment models**. accepted in Journal of Statistical Physics

2026 R. Hazra, R. van der Hofstad, R. Ray. **Percolation on preferential attachment models**. Electronic Journal of Probability, Volume 31 (2026).

2026 A. Garavaglia, R. Hazra, R. van der Hofstad, R. Ray. **Universality of the local limit of preferential attachment models**. accepted in Annals of Applied Probability.

2025 S. Banerjee, S. Bhamidi, R. van der Hofstad, R. Ray **Non-equilibrium coagulation processes and subcritical percolation on evolving networks**. ArXiv Preprint

2023 A. Cipriani, B. Dan, R. Hazra, R. Ray. **Maximum of the Membrane Model on Regular Trees**. Journal of Statistical Physics 190, Article number: 25(2023).

Industrial Experience

2023-2024 **Analysing CD behaviour in nano-lithography process** (Joint project with ASML)

Selected Talks

2026 Synergies between Geometry, Probability, and Computation in High Dimensions

2025 Probability and Statistics of Discrete Structures Seminar, SLMath (formerly MSRI)

2024 International Indian Statistical Association Conference, Cochin, India.

2023 INFORMS APS-23 conference, Nancy, France

2023 Mark-Kac Seminar, Utrecht University, The Netherlands

2022 Spring School: Random Geometric Graph, TU Darmstadt, Germany.

2021 NETWORKS Training Week, Asperen, The Netherlands

2021 CNRS-IRP Bézout-Eurandom meeting, Institut Henri Poincaré, Paris, France.

Positions of Responsibilities

2021–2022 **Tresurer** of 59-th board of **Panache E.S.B.V**

2018–2019 **Treasurer** for **Integration 2018** and **Integration 2019**, the Annual Techno-Cultural-Sports Fest of **Indian Statistical Institute, Kolkata**

2017-2018 **Treasurer** of Hostel Mess run by a Students Body

Technical Skills

Programming Languages: Python, C++

Software Packages: R ,LaTeX

Extra Curriculars

Hobbies: Travelling, Nature Photography

Activities: Drama, Football, Chess, Bridge

Languages

Bengali: Mother Tongue

Hindi: Native Language

English: 2nd Language